

Linear algebra – 4

Functions of matrices and matrix equations

Homework

12 сентября 2026 года

Theoretical minimum. Cayley–Hamilton reduces every power to the first n of them and writes A^{-1} as a polynomial in A . A function of a matrix is defined through the Jordan form or the Cauchy integral and depends only on the values of f and finitely many derivatives at the eigenvalues, so $\text{Spec } f(A) = f(\text{Spec } A)$. The equation $AX - XB = C$ is uniquely solvable exactly when the spectra of A and B are disjoint; the general principle is to regard a matrix equation as a linear map on the space M_n and to compute the spectrum of that map by the Kronecker product. Exponentials multiply only for commuting matrices, but AB and BA always share the characteristic polynomial. When an equation asks for a square root or a logarithm, the answer is dictated by the sizes of the Jordan blocks, and a nilpotent part usually makes the solution unique or non-existent rather than merely rare.

1. (Putnam 1994) For $n \geq 1$ let d_n be the greatest common divisor of the entries of $A^n - E$, where $A = \begin{pmatrix} 3 & 2 \\ 4 & 3 \end{pmatrix}$. Show that $\lim_{n \rightarrow \infty} d_n = \infty$.
2. (Vienna MC 2025) Let n be a positive integer and let $M_n(\mathbb{R})$ be the space of real $n \times n$ matrices. Consider $f: M_n(\mathbb{R}) \times M_n(\mathbb{R}) \rightarrow M_n(\mathbb{R}) \times M_n(\mathbb{R})$, $f(A, B) = (AB, BA)$.
 - a) Is f injective?
 - b) Is f surjective?
 - c) Find all pairs (A, B) of skew-symmetric real matrices which are fixed points of f .
3. (HFY 2018) A square matrix A of order n satisfies $A^2 = O$. Prove that $\text{rank } A \leq \frac{n}{2}$.

4. (IMC 2025) Denote by S the set of all real symmetric 2025×2025 matrices of rank 1 whose entries take the values -1 or $+1$. Let $A, B \in S$ be chosen independently and uniformly at random. Find the probability that A and B commute.
5. (Putnam 1985) Let G be a finite set of real $n \times n$ matrices $\{M_i\}$, $1 \leq i \leq r$, which form a group under matrix multiplication. Suppose that $\sum_{i=1}^r \operatorname{tr} M_i = 0$. Prove that $\sum_{i=1}^r M_i$ is the zero matrix.
6. (Putnam 2014) Let A be an $m \times n$ matrix with rational entries. Suppose that there are at least $m+n$ distinct prime numbers among the absolute values of the entries of A . Show that the rank of A is at least 2.
7. (МФТИ 2002) Let A and B be square matrices of order n . Prove that $P(A+B) = P(A) + P'(A)B$ for every polynomial P if and only if $AB - BA = B^2$.
8. (Mexmat, 2024) For two positive definite symmetric matrices prove the identity

$$\sqrt{A} \sqrt{\sqrt{A^{-1}} B \sqrt{A^{-1}}} \sqrt{A} = \sqrt{B} \sqrt{\sqrt{B^{-1}} A \sqrt{B^{-1}}} \sqrt{B},$$

where the square root of a positive definite symmetric matrix is the unique positive definite symmetric matrix whose square is the given one.

9. (Putnam 1997) For a positive integer n and a real number c define x_k recursively by $x_0 = 0$, $x_1 = 1$ and $x_{k+2} = \frac{cx_{k+1} - (n-k)x_k}{k+1}$ for $k \geq 0$. Fix n and take c to be the largest value for which $x_{n+1} = 0$. Find x_k in terms of n and k , $1 \leq k \leq n$.
10. Prove the Lie–Trotter formula and the Baker–Campbell–Hausdorff expansion up to the quadratic term, and deduce that $\det e^A = e^{\operatorname{tr} A}$ and that e^A is always invertible.
11. (Mexmat, 2001) Prove that for every positive integer n the matrix

$\begin{pmatrix} 2 & 1 & 1 \\ 0 & 2 & 1 \\ 1 & 1 & 1 \end{pmatrix}^n$ has two equal entries on the main diagonal.

12. (SEEMOUS 2017) Let $A, B \in M_n(\mathbb{R})$.
 - a) Prove that there is $a > 0$ such that for every $\varepsilon \in (-a, a)$ with $\varepsilon \neq 0$ the equation $AX + \varepsilon X = B$ has a unique solution $X(\varepsilon) \in M_n(\mathbb{R})$.
 - b) Prove that if $B^2 = E$ and A is diagonalisable, then $\lim_{\varepsilon \rightarrow 0} \varepsilon \cdot \text{tr}(BX(\varepsilon)) = n - \text{rank } A$.
13. (Romania national olympiad 2026) Let $A, B \in M_n(\mathbb{C})$ and let $a_0, a_1, \dots, a_m \in \mathbb{C}$ with $a_0 \neq 0$ and $a_m \neq 0$ satisfy $A(a_0E + a_1B + a_2B^2 + \dots + a_mB^m) = B$. Prove that $AB = BA$.
14. (IMC 2025 shortlist) Consider the differential equation $\frac{dY}{dx} = Y^2$ for $n \times n$ matrices. Does there exist a solution with $\det Y \neq 0$ defined on the whole real axis? Answer the question (i) for $n = 2024$ and (ii) for $n = 2025$.